Course Code	Course Title	Lecture			Semester: II
MTCS246PET	Big Data Analytics	L	T	P	
Version: 1.2	**Date of Approval:**	3	1	0	

Scheme of Instruction			Scheme of Examination		
No. of Periods	:	60 Hrs.	Maximum Score	:	100
Periods/ Week	:	4	Internal Evaluation	:	30
Credits	:	4	End Semester	:	70
Instruction Mode	:	Lecture	Exam Duration	:	3 Hrs.

Prerequisite(s): Computer Graphics

Course Objectives:

1. To learn the need of Big Data and the various challenges involved and to acquire Knowledge about different analytical architectures.
2. To understand Hadoop Architecture, ecosystems and acquire knowledge about the NoSQL database.
3. To acquire knowledge about the NewSQL, MongoDB and Cassandra databases.
4. To Apply the processing of Big Data with advanced architectures like Spark.

Course Outcomes (CO):

COs No.	Statement	Mapped Program Outcomes (POs)
CO_1	Understand the knowledge of Big Data, Data Analytics, challenges and their solutions in Big Data.	**PO_2, PO_3, PO_{12}**
CO_2	Analyze Hadoop Framework and Eco systems.	**PO_1, PO_2, PO_3, PO_4**
CO_3	Analyze MapReduce and Yarn, Work on NoSQL environment. MongoDB and Cassandra.	**PO_2, PO_3, PO_4, PO_9**
CO_4	Apply Big Data using Map-reduce programming in Hadoop and spark framework	**PO_2, PO_3, PO_4, PO_{12}**

PO_1- Engineering Knowledge, **PO_2**- Problem analysis, **PO_3**- Design/development of solutions, **PO_4**- Conduct investigations of complex problems, **PO_5**- Modern tool usage, **PO_6**- The engineer and society, **PO_7**- Environment and sustainability, **PO_8**- Ethics, **PO_9**- Individual or team work, **PO_{10}**- Communication, **PO_{11}**- Project management and finance, **PO_{12}**- Life-long Learning

Mapping of course outcomes with program outcomes

Course Outcomes	PO_1	PO_2	PO_3	PO_4	PO_5	PO_6	PO_7	PO_8	PO_9	PO_{10}	PO_{11}	PO_{12}
CO_1		2	1									3
CO_2	3	2	2	1								
CO_3		3	1	2					1			
CO_4		2	3	1								1

1 – Reasonable; 2 – Significant; 3 – Strong

Detailed Contents:

Unit: 1	Introduction to big data: Data, Characteristics of data and Types of digital data: Unstructured, Semi-structured and Structured - Sources of data. Big Data Evolution - Definition of big data-Characteristics and Need of big data-Challenges of big data. Applications of Big Data, Data Analytics, Types of Data Analytics, Data Analytics Methods and Techniques, Big data analytics, Data Lake, Architecture of Data Lake. Overview of business intelligence.
Unit: 2	Big data technologies and Databases: Hadoop – Requirement of Hadoop Framework - Design principle of Hadoop –Comparison with other system SQL and RDBMS- Hadoop Components – Architecture -Hadoop 1 vs Hadoop 2.
Unit: 3	MapReduce and YARN framework: Introduction to MapReduce, Processing data with Hadoop using MapReduce, Introduction to YARN, Architecture, Managing Resources and Applications with Hadoop YARN. Big data technologies and Databases: NoSQL: Introduction to NoSQL - Features and Types- Advantages & Disadvantages -Application of NoSQL.
Unit: 4	New SQL: Overview of New SQL - Comparing SQL, NoSQL and NewSQL. Mongo DB: Introduction – Features – Data types – Mongo DB Query language – CRUD operations – Arrays – Functions: Count – Sort – Limit – Skip – Aggregate – Map Reduce. Cursors – Indexes – Mongo Import – Mongo Export. Cassandra: Introduction – Features – Data types – CQLSH – Key spaces – CRUD operations – Collections – Counter – TTL – Alter commands – Import and Export – Querying System tables.
Unit: 5	Hadoop Frame Work: Map Reduce Programming: I/O formats, Map side join-Reduce Side Join-Secondary Sorting- Pipelining MapReduce jobs. Spark Frame Work: Introduction to Apache Spark-How spark works, Programming with RDDs: Create RDDspark Operations-Data Frame.

Examination and Evaluation Pattern: It include both internal evaluation (30 marks) comprising two class sessional exams/ assignments/ quiz/ seminar presentation etc. and external evaluation (70 marks) which is mainly end semester examination.

Text Books:

1 Tom White, "Hadoop: The Definitive Guide", O"Reilly, 4th Edition, 2015.